

A New Asymmetric Measure of Association for Ordinal Variables

Author(s): Robert H. Somers

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table of means (Table 3) suggests that the influence exerted on public affairs topics by the less educated opinion leaders may be colored by ideological orientations different from those among the better educated respondents. It would be of further interest to learn whether opinion leaders' sense of "conservatism" leads to differential selection of public affairs topics about which to exert influence. For instance, one might hypothesize that (controlling for education) the highly "resistant" opinion leaders confine themselves more to local issues, and the more "liberal" leaders exert influence on a more cosmopolitan range of topics. Similarly, "conservative" and "liberal" opinion leaders might be found to differ on level of information about public affairs topics (again, controlling for education).

It is somewhat disquieting to learn that for this sample high opinion leaders are as likely as the non-leaders to agree with statements expressing an anomic reaction to life, such as that "there's little use in writing public officials, because often they aren't really interested in the problems of the average man," that "nowadays, a person has to live pretty much for today and let tomorrow take care of itself" and that "in spite of what some people say, the lot of the average man is getting worse." This finding leads one to speculate (hopefully) that perhaps the two-item opinion leadership typology is too permissive or that the term "public affairs" is not a meaningful cue to respondents.

The interaction noted between age and sex on opinion leadership implies that leadership in the public affairs area is linked to respondents' life cycle. It may be that younger men actively pursuing careers and upwardly striving find that knowledge and influence concerning politics are instrumental and rewarding, while at the same time their wives are devoting energies to the domestic duties of raising families. Later in life, when, freed from home responsibilities, the women may find increasing opportunities and social rewards attendant upon opinion leadership.

Finally, it is not surprising that opinion leadership in public affairs is substantially correlated with education (.47) and with self-ascribed social class (.41).

A NEW ASYMMETRIC MEASURE OF ASSOCIATION FOR ORDINAL VARIABLES *

ROBERT H. SOMERS
The University of Michigan

A pair of asymmetric coefficients is introduced which are appropriate for measuring the association in ordered contingency tables. Earlier statistical methods are reviewed, and the new coefficients are shown to be closely related to both Kendall's tau-b and to Goodman and Kruskal's gamma, as well as the commonly used "percentage-difference." They are also shown to have a operational interpretation (in the Goodman-Kruskal sense). Their utility in both square and non-square contingency tables is discussed, as well as the way they may be interpreted as ordinal analogues of the traditional regression coefficients.

INTRODUCTION

IT is with some hesitation that I propose any new coefficients for measuring association in contingency tables, since the review by Goodman and Kruskal [1] suggests that any conceivable coefficient must have been considered by earlier workers. However, it appears that a very simple modification of two rather new coefficients both provides

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[1] Leo Goodman and William Kruskal, "Measures of Association for Cross Classifications, II," *Journal of the American Statistical Association,* 54 (March, 1959), pp. 123–163.

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analytically useful coefficients and suggests criteria for measuring "association" in tables of the type to be described below.

The two coefficients to be proposed, d_{xy} and d_{yx}, are identical in computation and interpretation, but they make a distinction between the independent and dependent variables in a bivariate distribution. As will be shown, they may be viewed as asymmetric versions of Goodman and Kruskal's gamma, and are also basic constituents of Kendall's tau. In keeping with product-moment convention, I shall use d_{yx} when X is the independent variable, and d_{xy} in the reverse situation.

Nothing in this paper treats the problem of estimation of such population measures from sample data, except for a brief note on testing the hypothesis that $d_{yx}=0$. The first task in any statistical analysis (hypothesis testing or other) is to determine the characteristics of the population that one wishes to describe in quantitative terms, and only later to derive the best estimation procedures from a randomly selected sample. It seems as though the greater need at the moment in sociology is the elaboration of more appropriate descriptive statistics (a task probably done better by sociologists than by statisticians) rather than the derivation of more precise inferential techniques for existing descriptive measures. In addition, the remarks to be made below about the origin of ordinal variables in sociology suggest some reasons why the question of sampling seems to require the prior solution of certain problems of measurement error, about which no adequate theory exists at present.

ORDINAL COEFFICIENTS AND MONOTONIC CORRELATION

It should be emphasized that the remarks contained in this paper relate specifically to *ordered* contingency tables, meaning that both variables have a natural ordering (i.e., are ordinal variables) like "status," "education," "degree of agreement," etc. They do not apply in situations where one or both variables are nominal, i.e., without a natural ordering.

Although it may be that some variables are inherently ordinal in nature (e.g., perceived social distance, where the perceiver makes only ordinal distinctions),[2] a more frequent basis for an ordinal variable seems to be situations in which the variable is considered an indicator of an "underlying" variable that is not directly observable, but is assumed to be monotonically related to certain manifest indicators. For example, "year of school completed" can be treated superficially as an interval scale in the sense that a person completing eight years of school can be said to have twice as much "schooling" as someone completing four years of school. In many situations, however, the intended meaning of "years of school completed" is to characterize the respondent in terms of an underlying continuum which might be described as "the amount of knowledge, understanding, and general educational skill acquired," which is in reality a complex of microsociological stimuli that would be virtually impossible to measure directly. For purposes of analysis, the easily available indicator "years of school completed" seems to be a good approximation, but cannot be assumed to be an interval measurement of the underlying variable. In other circumstances, some kind of general education examination or test might be chosen as an alternative indicator variable. Treated as *ordinal* variables, the relation of either "indicator variable" to a third should be about the same.[3] These considerations seem to the present writer to be an important rationale for the treatment of some variables as ordinal even though they seem, superficially, to be interval scales—a rationale that is especially pertinent to comparative research, where one may, of necessity, use different "indicator" variables.

Whatever the assumed origin of the ordinal variable, its chief characteristic is that one cannot think in terms of comparable units of measurement, but only an ordering. In the example above, one would say that a person with eight years of schooling has more education than one with four years, but one would not say this is "twice as much educa-

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[2] Florian Znaniecki, *The Method of Sociology*, N. Y.: Rinehart, 1934, p. 309.

[3] This seems to be the same point as is made by Lazarsfeld's "interchangeability of indices" notion. See Paul Lazarsfeld and Morris Rosenberg, editors, *The Language of Social Research*, Glencoe, Ill.: The Free Press, 1955, pp. 73–77.

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tion." This is obviously a very weak kind of measurment, and ways of strengthening it without going to full interval measurement might be considered. (There are certain indications, for example, that having bivariate rather than univariate data in ordinal measurement makes the measurement somewhat less weak; these ideas will be discussed in a subsequent paper.)

The cross classification of two ordinal variables leads directly to the notion of monotonic correlation that has recently been discussed by Burr,[4] and is implicit in the work of Kendall and Goodman and Kruskal. Perfect monotonic correlation is represented by a situation in which, for two variables X and Y, the value of X increases as Y increases, and conversely, regardless of the (possibly changing) rate of increase. Thus linear correlation is a special case of monotonic correlation; in general, it refers to the notion of a curvilinear relationship that does not "double back" on itself. Put more formally, for each X there is a unique value of Y, and conversely.

When working with categorical data it is useful to consider a modified form of monotonic correlation which permits a zero slope, but no doubling back: as X increases, Y never decreases. To distinguish the first type from this more general case, we shall call the first, where no zero slope is permitted, "strictly monotone."

PRESENT NOTIONS OF ASSOCIATION

Although the very useful notion of an "operational interpretation" of a measure of association, introduced by Goodman and Kruskal,[5] leads to the suggestion that the merits of a measure of association cannot be evaluated independently of its research context, it is nevertheless true that comparability of different pieces of social research would be enhanced by a rather general "work-horse" measure of association with an interpretation appropriate for the fairly general type of research where one is simply looking for relationships which might possibly be causal in nature.

In many instances it is sufficient simply to present a percentaged table, percentaged in the direction of the independent variable,[6] so that estimates of conditional probabilities in the population may be examined for the magnitude of their difference. In fourfold tables this procedure is completely unambiguous. In tables that are larger than 2 x 2 this simple criterion becomes somewhat intuitive, since there is more than one pertinent percentage difference to compare. In ambiguous situations, it becomes easy for the analyst to emphasize those particular differences that confirm his preconceptions.

To solve such problems of objectivity the investigator frequently has had recourse to a χ^2 analysis. As a general tool of statistical *inference* in situations appropriate to it, such an analysis is completely unobjectionable. Yet its ready availability has sometimes resulted in inappropriate usage.[7]

For example, it is a tool of inference (in situations of simple random sampling) and is often construed as a technique for describing something about the sample, rather than the population. In part, this is because it does not derive from a clearly conceived population parameter as does, for example, a sample correlation coefficient. Instead, it is a way of using a sample of observations to test the inference of statistical independence in the population, and the function of sample values that is used in making this test is not a useful descriptive statistic.

Second, one sometimes uses descriptive coefficients derived from the sample value of χ^2, such as Tschuprow's or Cramer's T, but overlooks the fact that they are not sensitive to the ordering of categories that is inherent in the nature of ordinal variables. Hence they measure *any* deviation of sample observations from statistical independence, not just a deviation in the direction of monotonic correlation, which may be the hypothesis of interest with ordinal variables.

[4] Edward J. Burr, "Distribution of Kendall's Score S for a Pair of Tied Rankings," *Biometrika,* 47 (June, 1960), p. 151.

[5] Leo Goodman and William Kruskal, "Measures of Association for Cross Classifications, I," *Journal of the American Statistical Association,* 49 (December, 1954), p. 740.

[6] Hans Zeisel, *Say It with Figures,* N. Y.: Harper, 1957, Chapter 2.

[7] Philip J. McCarthy, *Introduction to Statistical Reasoning,* N. Y.: McGraw-Hill, 1957, Chapter 11 is probably the best discussion of χ^2 procedures for the introductory student. Goodman and Kruskal, *op. cit.* (1954), also discuss this point.

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Third, because of this insensitivity to ordering, the χ^2 test used as an inferential tool is less powerful against population hypotheses of monotonic correlation than would be a test designed with such hypotheses specifically in mind. Thus when some monotonic correlation exists in the population the traditional χ^2 test is less likely to reject the null hypothesis than, for example, a test for the significance of Kendall's S, which is designed specifically for such alternative hypotheses.[8]

To fill the gap left by such techniques, Kendall[9] and Goodman and Kruskal[10] have proposed, among other measures, two which have found wide usefulness: Kendall's tau and Goodman and Kruskal's gamma. For a full discussion the reader is referred to the original literature; for the present discussion a brief review will be given.

Kendall's tau has a long history;[11] basically, it derives from the notion of correlating signs of differences between all possible *pairs* of observations; for this reason it is sometimes called the coefficient of sign correlation. Originally proposed for the case of fully ranked data (which would lead to ordered contingency tables with only one entry per row and column), it was subsequently modified for the case of a partial ranking of both variables (i.e., tied ranks were permitted) and so became applicable to what we call ordered contingency tables. For such cases Kendall suggested three possible coefficients, which he designated tau-a, tau-b, and tau-c. Each of these is, like an ordinary correlation coefficient, and unlike a percentage difference, symmetric in its treatment of the variables, so that no distinction is made between the independent and the dependent variable. The three possible versions of tau all have the same numerator, which we shall call (following Kendall) P—Q=S, the excess of concordant over discordant pairs of observations. For the computation of P and Q, which is difficult to express in symbolic form, the reader is referred to Kendall[12] where it is described in words. No operational interpretation of tau-b or tau-c is available, which means that one cannot express in words (in terms of probabilities, errors in prediction, etc.) the meaning of a value of tau equal to .6, for example. Tau-b, which is the most widely used measure of the three versions of tau for the case of tied ranks, achieves an extreme of ± 1.0 when all frequencies fall in the diagonal. For a table that is not square this maximum cannot be achieved, and a minor modification is introduced to correct for this, leading to tau-c. Although this last version of tau thus has the virtue of flexibility over different table formats, it has no particular interpretation for values intermediate between 0 and 1.0 and, of course, treats the variables symmetrically.

A significant advance was made by Goodman and Kruskal in introducing the notion of operational interpretation, which provides, in effect, a formal criterion of "association" for tables intermediate between the extremes of range of a coefficient. Along with several other measures for nominal variables, they introduced gamma for ordinal variables. This measure, like tau, has numerator P—Q; clearly, all of these coefficients measure essentially "the same thing," but may differ rather radically in value (as will be illustrated below) because of their different denominators. Gamma has the virtue of an interpretation: it tells us the proportionate excess of concordant over discordant pairs among all pairs which are fully discriminated, or fully ranked. Thus it omits from consideration pairs which are tied on either X or Y, or both. The denominator is, in fact, simply P + Q, and gamma can be thought of as a generalization of Yule's Q for tables larger than 2 x 2. Like the various versions of tau, gamma treats the variables symmetrically.

As a final criterion of association which will enter into the ensuing discussion, it should be pointed out that some coefficients, especially those appropriate to nominal vari-

[8] Burr, *op. cit.*, p. 156, exemplifies a sample yielding significant S, non-significant χ^2.

[9] Maurice G. Kendall, *Rank Correlation Methods*, N. Y.: Hafner, 1955, Chapter 3.

[10] Goodman and Kruskal, *op. cit.* (1954), pp. 747–754.

[11] William Kruskal, "Ordinal Measures of Association," *Journal of the American Statistical Association*, 53 (December, 1958), p. 854.

[12] Kendall, *op. cit.*, p. 46.

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ables, utilize the notion of categorical prediction.[13] Such coefficients measure the extent to which a knowledge of a respondent's location on X, say, helps in predicting his location on Y. This criterion is often useful, but does not, by itself, take into account the notion of ordering.

THE PROPOSED MEASURES

The coefficients proposed in this paper will be called d_{xy} and d_{yx}. They are asymmetric: one would use d_{xy} when a distinction is called for between Y as the independent variable and X as the dependent, and d_{yx} when the role of the variables is reversed. Again, these coefficients use P—Q in the numerator, and so they differ from the measures discussed above only in terms of the denominator. The denominator of d_{yx} is the number of pairs not tied on X (without regard for Y), while the denominator of d_{xy} is the number of pairs not tied on Y (without regard for X). These concepts will be clarified below.

In addition to the virtue of asymmetry, which is useful in many situations, they have an operational interpretation like gamma, so that one may describe exactly what is meant by intermediate values of a coefficient. Finally, these measures are closely related to the percentage difference. In fact, as will be shown below, they are equivalent to the percentage difference in a fourfold table, and may be seen as a function of certain percentage comparisons in a table of larger size. Again, it should be emphasized that they are, like gamma and tau, appropriate only to situations where both variables have a natural ordering and one wishes to measure the extent to which there is some form of monotonic correlation. This will be made clear in illustrative material given below. Also, like gamma and tau they measure only the extent to which there is a like ordering between the variables, without taking distance into account.

THE NOTION OF PAIRS OF OBSERVATIONS

In order to discuss intelligibly the basis for the proposed coefficients, some introduction must be given to the notion of pairs of observations. As Goodman and Kruskal[14] point out, when dealing with ordinal variables it is sometimes very useful to think in terms of certain simple characteristics of the distribution of pairs of observations, rather than the individual observations themselves. A frequency distribution over two ordinal variables, such as Table 1 (where we arbitrarily take a 2 x 3 table for illustration), can be simply translated into a frequency distribution over pairs of observations, as in Table 2, no matter what the format or size of the table. In this way the problem of generalizing coefficients devised for a 2 x 2 table, for example, can be solved immediately.

Although the example given in Tables 1 and 2 is for a 2 x 3 table, it should be made clear that analogous computations can reduce any ordered contingency table to a frequency distribution of pairs over the (mutually exclusive and exhaustive) set of categories shown there. Moreover, all measures of an ordinal nature seem to derive

TABLE 1. HYPOTHETICAL FREQUENCY DISTRIBUTION FOR TWO ORDINAL VARIABLES

	X_1	X_2	X_3	
Y_1	a	b	c	B_1
Y_2	d	e	f	B_2
	A_1	A_2	A_3	n

TABLE 2. DISTRIBUTION OF DISTINGUISHABLE PAIRS, FOR FREQUENCIES IN TABLE 1

Type of Pair	Number of Pairs	Symbolic Designation
Concordant	$a(e+f)+b(f)$	P
Discordant	$c(d+e)+b(d)$	Q
Tied on X only	$ad+be+cf$	X_o
Tied on Y only	$a(b+c)+bc+d(e+f)+ef$	Y_o
Tied on both X and Y	$(1/2)\{a(a-1)+b(b-1)+c(c-1)+d(d-1)+e(e-1)+f(f-1)\}$	Z
Total	$(1/2)\{n(n-1)\}$	$\binom{n}{2}$

[13] Goodman and Kruskal, *op. cit.* (1954), p. 740 and p. 759.

[14] *Ibid.*, p. 748.

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from such a distribution of pairs. For example, as was noted earlier,

$$\text{gamma} = \frac{P-Q}{P+Q} \text{ ; moreover,}$$

$$\text{tau-a} = \frac{P-Q}{\binom{n}{2}},$$

$$\text{tau-b} = \frac{P-Q}{\sqrt{(P+Q+X_o)\,(P+Q+Y_o)}}, \text{ and}$$

$$\text{tau-c} = (P-Q)\,\frac{2m}{n^2(m-1)},$$

where m represents the number of rows or columns in the table, whichever is smaller. Finally, the proposed coefficients are

$$d_{yx} = \frac{P-Q}{P+Q+Y_o}, \text{ and}$$

$$d_{xy} = \frac{P-Q}{P+Q+X_o}.$$

Since the denominator terms appear in both Kendall's tau-b and in d_{yx} and d_{xy} in the same way, it is convenient to have a simpler notational designation for them. Let $P+Q+X_o = Y_u$, and $P+Q+Y_o = X_u$; then $d_{yx} = (P-Q)/X_u$, and $d_{xy} = (P-Q)/Y_u$. This form of notation is chosen because the computation for X_u gives the number of *pairs of observations that are not tied on X,* regardless of their status on Y. Similarly, Y_u is the number of pairs untied on Y, without regard for X. With the marginal totals shown in Table 1, it is simpler to compute the value of X_u and Y_u as follows:

$$X_u = A_1A_2+A_1A_3+A_2A_3, \text{ and}$$
$$Y_u = B_1B_2.$$

In general the computation of the number of untied pairs on one variable has reference only to the (univariate) marginal distribution, and is a sum of products of marginal frequencies, two at a time. In some situations it may be simpler to compute $X_u = (\frac{1}{2})(n^2-\Sigma A_j^2)$, where the summation is over all column totals, and Y_u analogously from the row totals.

It is clear from the above definition of the new coefficients that they differ only slightly from either Kendall's tau-b or Goodman and Kruskal's gamma. In fact, $(\text{tau-b})^2 = d_{xy}d_{yx}$, i.e., the d's bear the same relation to Kendall's correlation analogue as the classical regression coefficients bear to the product-moment correlation ($r^2 = b_{xy}b_{yx}$). They may therefore be spoken of as *analogues of the ordinary regression coefficients,* but this analogy is not so useful in their interpretation. The notion of slope, in fact, has little meaning in relation to monotonic correlation. In addition, the numerator and denominator separately may be thought of as "ordinal analogues" of the covariance and variance, respectively, and they are precisely the covariance and variance (when divided by n^2) of the distribution over *pairs* of observations that is formed by assigning a +1, −1, or 0 for pairs in accord with, discordant with, or tied with respect to some *a priori* ordering.[15]

The most useful interpretation of the coefficient d_{yx} follows closely the interpretation of gamma.[16] It expresses the proportionate excess of concordant pairs over discordant pairs among pairs not tied on the independent variable. Put differently, our d_{yx} is the difference between conditional probabilities of like and unlike order, under the condition that we ignore ties on the independent variable (although they will be present, in general, on the dependent variable). (Note that in order to speak of probabilities here we must think in terms of drawing pairs at random from the bivariate frequency distribution.)

THE RELATION OF d_{yx} TO THE PERCENTAGE DIFFERENCE

Because both d_{yx} and the percentage difference are asymmetric measures of relationship, it is reasonable to ask whether they are related, and if so, in what way. To study this question we shall consider four cases separately: (a) 2 x 2 or fourfold tables, (b) 2 x c tables, i.e., tables where the dependent variable is dichotomous, (c) r x 2 tables, where the independent variable is dichotomous, the dependent polytomous, and finally (d) r x c tables, where both r and c exceed 2. In all but the first case (fourfold tables)

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[15] Kendall, *op. cit.,* Chapter 2.

[16] Goodman and Kruskal, *op. cit.* (1954), p. 753. See also the discussion of gamma (γ) in W. Allen Wallis and Harry V. Roberts, *Statistics: A New Approach,* Glencoe, Ill.: Free Press, 1956, pp. 282–284.

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TABLE 3. HYPOTHETICAL DISTRIBUTION IN FOURFOLD TABLE FORMAT

	X_1	X_2	Difference
Y_1	55.8%	13.0%	42.8%
Y_2	44.2	87.0	−42.8
Total	100.0% (113)	100.0% (23)	0.0
$d_{yx}=\frac{1110}{2599}=.427$			tau-a=.121
			tau-b=.320
$d_{xy}=\frac{1110}{4620}=.240$			tau-c=.240
			gamma=.787

the meaning of percentage difference is of course not clear. We shall show that in a fourfold table d_{yx} is equal to the percentage difference, and in larger tables, under certain circumstances, d_{yx} can be viewed as a function of certain percentage differences.

Fourfold Tables. To see the relation between d_{yx} and the percentage difference in a fourfold table (in which the frequencies are designated $\begin{array}{c|c} a & b \\ \hline c & d \end{array}$), note that the percentage difference is

$$100\left(\frac{a}{a+c}-\frac{b}{b+d}\right)=100\frac{ad-bc}{(a+c)(b+d)}=100\,d_{yx}.$$

In Table 3 is shown a fourfold table in percentaged form, together with the values of d_{yx}, d_{xy}, and the other coefficients discussed here. Note that, because of the skewness of the marginals (i.e., few untied pairs on X), the value of gamma is much larger than that of the other coefficients. Also, tau-b is the geometric mean of d_{yx} and d_{xy}, and is intermediate in value.

Polytomous Independent Variable. In tables of larger size the situation is somewhat more complicated, and we start with the case of a dichotomous dependent variable with polytomous independent variable (2 x c format). In Table 1, for example, there are three possible fourfold tables, obtained by comparing the first two columns, the first and last, and the last two. In the accompanying diagram we have designated these Tables 4A, 4B, and 4C. The coefficient d_{yx} for Table 1 can be seen as a simple weighted average of the d's in these three tables, where the weights are the denominators of the individual d's.

Below Tables 4A-C we show the computation of the d's for each sub-table of Table 1. Each of these d's is of course the percentage difference in that table. For the first subtable, for example, we have $d_A=(ae-bd)/A_1A_2$. To obtain the value of d_{yx} for the composite Table 1, we simply form a weighted average for these separate d's, where the weights are the proportion of pairs not tied on X in each sub-table, which is equivalent to weighting the d's by their denominators.

In general, when the dependent variable is dichotomous, nothing new arises and the coefficient may be interpreted as a simple weighted average of the percentage differences in the several fourfold tables of which the actual tabulation is a composite, which are obtained by considering each of the $\binom{c}{2}$ distinguishable pairs of columns.

Polytomous Dependent Variable. When we turn from a dichotomous dependent variable to one that is polytomous, the situation is a good deal more complex. Indeed, the simplest possible case of this type, where we have a 3 x 2 table, presents crucial difficulties in measuring order relations, as will be seen. The traditional χ^2 methods ignore, of course, the fact that our trichotomy is

TABLE 4. SUB-TABLES ARISING FROM TABLE 1

4A

	X_1	X_2
Y_1	a	b
Y_2	d	e
	A_1	A_2

$d_A=\frac{ae-bd}{A_1A_2}$

4B

	X_1	X_3
Y_1	a	c
Y_2	d	f
	A_1	A_3

$d_B=\frac{af-cd}{A_1A_3}$

4C

	X_2	X_3
Y_1	b	c
Y_2	e	f
	A_2	A_3

$d_C=\frac{bf-ce}{A_2A_3}$

$d_{yx}=W_1d_A+W_2d_B+W_3d_C$

where $W_1=A_1A_2/W.$
$W_2=A_1A_3/W.$
$W_3=A_2A_3/W.$
$W.=A_1A_2+A_1A_3+A_2A_3$

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TABLE 5A. 5 x 2 TABLE SHOWING PERCENTAGE DIFFERENCES

	Frequencies		Proportions		Differences
	X_1	X_2	X_1	X_2	
Y_1	8	2	.40	2/17= .12	+.28
Y_2	6	5	.30	5/17= .29	+.01
Y_3	3	4	.15	4/17= .24	−.09
Y_4	1	3	.05	3/17= .18	−.13
Y_5	2	3	.10	3/17= .18	−.08
Total	20	17	1.00	1.01	−.01

P	−Q
120	−24
60	−30
18	−12
3	−6
201	−72=129

$$d_{yx}=\frac{129}{340}$$

ordered. Perhaps a better alternative than χ^2 is to assume an interval scale and assign numbers, e.g., 0, 1, 2 to the three ordered categories. This has been suggested as one way of strengthening the common χ^2 test.[17]

As an alternative to the assumption of an interval scale, however, the nature of the data may suggest a coefficient appropriate to ordinal variables. In particular, the d_{yx} proposed in this paper is probably the most sentitive to percentage differences because of its asymmetry.

To illustrate the manner in which d_{yx}, when applied to an r x 2 table, takes into account both ordering and the differences in percentages, we shall deal with a 5 x 2 table, such as that shown in Table 5A. However, it must first be pointed out that this interpretation of d_{yx} in terms of percentage differences has been made only for r x 2 tables in which the distribution of frequencies exhibits a certain type of monotonicity. This type of monotonicity may be described by saying the table has only one "transition point." By a transition point is meant a change in sign as one reads down the column of percentage differences. It will be noted that Table 5A is of this character. This can be seen as a type of monotonicity if one notes that, in this example, the X_2's are underrepresented in all the Y-categories above the transition point, and over-represented in all the Y-categories below that point. Two additional facts should be noted. First, the actual magnitude of the percentage difference is of no consequence to the notion of transition point, but only its sign. Second, the concept of transition points does not apply unless the dependent variable contains more than two categories. It would have no meaning, for example, in a fourfold table.

With this qualification, the manner in which d_{yx} is a function of percentage differences in this special type of r x 2 situation can be illustrated. Consider what intuitive notion of association might be used to charac-

TABLE 5B. 5 x 2 TABLE COLLAPSED AROUND TRANSITION

	X_1	X_2
Y_1, Y_2	14	7
Y_3, Y_4, Y_5	6	10
	20	17

$$\frac{\%\text{ difference}}{100}=\text{PD}=\frac{140-42}{340}=\frac{98}{340}=.29$$

$$\delta_1=\left(\frac{8}{20}\right)\left(\frac{5}{17}\right)-\left(\frac{2}{17}\right)\left(\frac{6}{20}\right)=\frac{28}{340}\quad\text{in upper half.}$$

$$\delta_2=\left(\frac{3}{20}\right)\left(\frac{3}{17}\right)-\left(\frac{4}{17}\right)\left(\frac{1}{20}\right)=\frac{5}{340}$$

$$\delta_3=\left(\frac{3}{20}\right)\left(\frac{3}{17}\right)-\left(\frac{4}{17}\right)\left(\frac{2}{20}\right)=\frac{1}{340}\quad\text{in lower half.}$$

$$\delta_4=\left(\frac{1}{20}\right)\left(\frac{3}{17}\right)-\left(\frac{3}{17}\right)\left(\frac{2}{20}\right)=\frac{-3}{340}$$

$$\sum_i\delta_i=\qquad\qquad=\frac{31}{340}$$

$$d_{xy}=\text{PD}+\sum_i\delta_i=\frac{98}{340}+\frac{31}{340}=\frac{129}{340}$$

[17] William G. Cochran, "Some Methods for Strengthening the Common χ^2 Tests," *Biometrics*, 10 (December, 1954), p. 434.

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terize the degree of association in a table such as Table 5A. One such notion might be to use the magnitude of the percentage differences, row by row. If we take a simple total, this is of course zero (apart from rounding error). Alternatively, we could take the sum of the absolute values. If instead we take merely the sum of the positive (or negative) differences we get half the sum of the absolute values. Finally, in tables like this, with only one transition, we can get the same information by collapsing the polytomous attribute around the transition point, as is shown with Table 5B. The result is a fourfold table with a percentage difference equal to the sum of the differences with positive (or negative) signs.

Thus we could characterize the association in this way just as well by collapsing to a fourfold table. But we would, of course, have lost a certain amount of information that would be contained in the collapsed categories. The coefficient d_{yx} does not lose this information, but contains an additional element, as follows.

Each of the collapsed portions of the table can be considered as made up of one or more fourfold tables prior to collapsing. Thus the lower half of the collapsed table contains three fourfold tables obtained by taking successively rows 3 and 4, rows 3 and 5, and rows 4 and 5. The upper half of the collapsed table, on the other hand, contains only one such table, making four altogether.

The manner in which d_{yx} utilizes the information from these collapsed portions of the table is to add up the $\delta = ad - bc$ value for each of the tables, as is shown in Table 5B. Note that these δ's are computed directly on the proportions. (In Table 5B we have retained fractions to avoid rounding error.) These δ's are greater as the ordered pattern of association is more prominent, i.e., as the percentage differences increase (in absolute value) in moving away from the transition point. The overall coefficient reaches a maximum when there is complete separation at the transition, so that the collapsed fourfold table shows perfect association (see below). Finally, the value of d_{yx} is simply the sum of these δ-values, plus the percentage difference (PD) in the collapsed table, as is shown at the bottom of Table 5B.

It is in this way that d_{yx} can be seen as a function of the percentage differences in the more complex situation where the ordering is over a polytomy rather than a dichotomy. For r x 2 tables in general, the same interpretation of d_{yx} can be made, so long as the table contains only one transition point. So far nothing has been said of tables with more than one transition point, and we shall defer a consideration of such cases until after the next step, which involves a consideration of the general r x c ordered contingency table.

General r x c Tables. In the above discussion we have examined the relation between d_{yx} and the percentage differences in 2 x 2, 2 x c, and r x 2 ordered contingency tables, and have provided the necessary background for a consideration of the r x c table. Nothing new, in fact, is added in this general case. From the 2 x c case comes the suggestion that we should consider each pair of columns of the table separately, and quantify each such comparison. The value of d_{yx} for the whole table can be expected to be a weighted average of such pair-wise comparisons, where the weights are the products of column marginals, and each pair of columns is characterized by its own d_{yx} value. From the r x 2 case we have found how to quantify the order relationship when r exceeds 2. The whole procedure is illustrated in Table 6, where we provide a numerical example of a 3 x 3 table for simplicity. To avoid the complexities of algebraic notation, we shall consider the interpretation of d_{yx} as a function of all possible ordered percentage comparisons (in tables that are monotonic in the above sense) concluded with the presentation of Table 6.

We have noted earlier that the illustrated method for computing d_{yx} from percentage comparisons in the r x 2 situation requires that the data exhibit the type of monotonicity leading to one (and only one) transition point. Rules can be devised for computing d_{yx} from a percentaged table in the more general situation where more than one transition point exists, but the illustrated rules should not be applied to data that show more than one transition point, i.e., that deviate from this type of monotonicity. These rules, at any rate, are

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TABLE 6. ILLUSTRATION OF COMPUTATION OF d_{yx} FROM PERCENTAGES IN 3 X 3 ORDERED CONTINGENCY TABLE

Proportions

	X_1	X_2	X_3
Y_1	.7	.4	.2
Y_2	.2	.3	.3
Y_3	.1	.3	.5
Number of cases	(20)	(30)	(60)

	X_1	X_2
Y_1	.7	.4
Y_2	.2	.3
Y_3	.1	.3
	(20)	(30)

PD=.30
δ=.03
d_{12}=.33

	X_1	X_3
Y_1	.7	.2
Y_2	.2	.3
Y_3	.1	.5
	(20)	(60)

PD=.50
δ=.07
d_{13}=.57

	X_2	X_3
Y_1	.4	.2
Y_2	.3	.3
Y_3	.3	.5
	(30)	(60)

PD=.20
δ=.06
d_{23}=.26

Weights (20)(30)=600 (20)(60)=1200 (30)(60)=1800

$$d_{yx}=\frac{6(.33)+12(.57)+18(.26)}{6+12+18}=\frac{13.5}{36}=.375$$

not intended as computing rules, but rather as a means for providing an interpretation of the value of d_{yx} by showing how it relates to percentage comparisons.

As a result of this qualification, the interpretation of d_{yx} for the general r x c table applies only when each pair of columns, considered separately, exhibits only one transition point, where the whole table, that is, exhibits a certain form of monotonicity that is closely related to the isotropy discussed by Yule and Kendall.[18] Although rules exist which relate d_{yx} to percentage comparisons in tables that do not exhibit this type of monotonicity, it is perhaps better to think of d_{yx} as pertaining to bivariate distributions that are of this sort, in the same sense that the traditional correlation and regression techniques can be interpreted, strictly, only in distributions where the regressions are linear. In this sense, this notion of monotonicity appears to play the same role in the interpretation of these coefficients as the notion of linearity plays in traditional correlation theory.

[18] G. Udny Yule and Maurice G. Kendall, *An Introduction to the Theory of Statistics*, London: Griffin, 14th edition, 1953, p. 57.

A COMPARISON OF d_{yx} AND GAMMA

A concluding note on the similarity between d_{yx} and gamma is in order. It is well known that one sometimes gets quite different results according to which way one runs the percentages, and an asymmetric coefficient is therefore inherently more sensitive to the "percentage" criterion of association. Nevertheless, gamma, although it is symmetric, shows a surprising sensitivity to percentage differences. For example, in r x 2 tables it is easy to show that neither d_{yx} nor gamma changes as the marginal distribution on the independent variable (the dichotomy) changes, as long as the percentage differences remain the same. On the other hand, for 3 x 3 and larger tables, both d_{yx} and gamma may change, and in general do change, when one marginal distribution is varied, despite the fact that the percentage distributions within the columns remain the same. This is a reasonable behavior for such coefficients to exhibit since they reflect not only percentage differences but also the weights that derive from the frequencies in the various columns.

The important difference between these two coefficients is that they actually measure quite different types of association. For a

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square table, d_{yx}, like tau-b, requires that all the frequencies fall in the main diagonal in order to achieve 1.0. (It should be noted, incidentally, that this is not the equivalent of a linear relationship, which is meaningless with ordinal measurement.) For a nonsquare table, d_{yx} requires a monotonic pattern. In both cases, gamma reaches a maximum under a wider class of situations, and so in general it is larger in magnitude than d_{yx}. Although this difference between gamma and d_{yx} is obvious in fourfold tables (cf. Table 3), it appears that (for constant sample size) the difference decreases as the number of rows and columns increases. This requires further investigation.

As a final interpretation of d_{yx}, it is useful to note that it is simply gamma modified by a penalty for the number of pairs tied on X only. This number of pairs is added to the denominator of gamma before taking the ratio, and as has been shown elsewhere, the "number of pairs tied on one variable only" is indentical to the expected frequency of error in a certain type of categorical prediction.[19]

It may appear to many readers that a discussion of the behavior of coefficients is rather "academic," and that they exhibit only small differences that are of little consequence. As Table 3 shows, however, the differences in value may not always be small. In addition, for those concerned with comparability of research some of these differences may assume crucial importance.

One question that relates to this discussion is the extent to which the relationship as measured by a particular coefficient is stable over various choices of "cutting points," where the bases of categorization are arbitrary. The question is impossible to answer without specifying in some way the form of the bivariate distribution under consideration. However, it does seem undesirable to have a coefficient that increases markedly in value when a table is collapsed. Gamma will exhibit such behavior on occasion, to a greater extent then Kendall's tau-b,[20] apparently because gamma gives no consideration to tied pairs. The behavior of d_{yx} has not been studied systematically in this connection but seems better because some tied pairs are included in its computation. This is also indicated by the earlier described relation between d_{yx}, d_{xy} and tau-b, together with the fact that tau-b is more "stable" in this sense.

PERCENTAGE DIFFERENCE VS. PREDICTABILITY

Tables that are not square introduce certain problems in the use of these asymmetric coefficients. In particular, it can be argued that in some such situations the criterion of ordinary percentage difference is not the most appropriate, but that one should instead refer to the notion of predicatability. In effect, when dealing with the proposed coefficients, this means referring to percentage differences that are run in the direction of the *dependent variable,* rather then in the usual direction of the independent variable. This parodoxical situation can be better understood by referring to the simplest example of a 3 x 2 table, such as Table 7A.

TABLE 7A. HYPOTHETICAL DISTRIBUTION WHERE $d_{yx}=1.0$

	X_1	X_2	
Y_1	25	0	25
Y_2	85	0	85
Y_3	0	30	30
	110	30	

$$d_{yx}=\frac{3300}{3300}=1.0$$

$$d_{xy}=\frac{3300}{5425}=.61$$

TABLE 7B. HYPOTHETICAL DISTRIBUTION WHERE $d_{yx}<1.0$

	X_1	X_2
Y_1	25	0
Y_2	0	30
Y_3	85	0
	110	30

$$d_{yx}=\frac{1800}{3300}=-.55$$

[19] Robert Somers, "A Similarity between Goodman and Kruskal's tau and Kendall's tau, with a Partial Interpretation of the Latter," ditto, November, 1961, to be published.

[20] This has been independently noted, at least by the present writer and Philip Converse (personal communication).

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Suppose that in this table, as before, X is the independent and Y the dependent variable. There is only one transition, as the term is used above, and indeed there is perfect separation between X_1 and X_2 above and below the transition point. Hence the value of d_{yx} is 1.0, a maximum. In terms of the "percentage difference plus ordering criterion," the association is perfect. (Had there been two transition points, but still perfect separation as one crossed the transitions, as in Table 7B, then the percentage difference criterion would indicate perfect association, but the ordered pattern is no longer present, so d_{yx} is less than 1.0.)

Suppose, however, that we ask for a somewhat different criterion of maximum association, viz., one that tells us something about the extent to which we can *predict* from the independent variable to the dependent. In Table 7A it is quite clear that knowledge of a respondent's X-category does not enable us to predict his Y-category with certainty. Knowing that he is in X_2 provides sure knowledge that he is in Y_3, but a knowledge that he is in X_1 does not provide such certainty.

Consequently, a measure that is based on predictive ability plus ordering would not be a maximum in such a situation. Indeed, as long as we are predicting *from few categories to many,* as in this situation, it is obviously unreasonable to ask for accurate prediction.

In such situations one may desire to measure predictability plus order instead of using d_{yx}. The proposed solution for such a situation is, surprisingly, to use d_{xy}. Its relevance can be illustrated by referring again to Table 7A. The value of d_{xy} in this table tells us the extent to which we have both predictability and ordering. Its magnitude, $d_{xy}=.61$, represesents the maximum that can be attained in such a situation where we are predicting from two categories to three (with these frequencies).

On the other hand, still referring to Table 7A, suppose that we take Y as the independent variable. We are then predicting from three categories to two, and could reasonably have perfect prediction. If we have both perfect predictability and perfect ordering, as here, then d_{yx} is 1.0. The full paradox is realized by noting that in such a situation, where Y is the independent variable, the use of d_{yx} may be interpreted as percentaging in the direction of the dependent variable, opposite to the traditional (and logical) procedure.

This paradox suggests that there is something contradictory about the notion of percentage differences and the notion of predictability, at least when order-pattern is an additional criterion in both cases. The full nature of this paradox has not been explored, but it is interesting to consider whether such a seeming paradox arises in traditional product-moment theory. A little thought will show that perfect prediction in one direction but imperfect prediction in the reverse direction, with interval scales, implies that there is nonlinearity of regression. Hence we might examine the correlation ratio, which is a useful decriptive device for nonlinear situations.

In this case the same distinction arises, since the correlation ratio for the regression of Y on X, for example, is defined as the proportion of variance of the *dependent* variable that is explained by the independent. Thus the variance of the dependent variable goes into the denominator of the correlation ratio, while the variance of the *independent* variable is the denominator of the regression coefficient. Hence one might say that in computing d_{xy} when X is the independent variable, and therefore when computing and averaging percentages run in the direction of the dependent variable, one is performing an operation on ordinal variables that is analogous to a computation of the correlation ratio for interval variables. The analogy is not strict, however, since in the computing of d_{yx} we are using the analogue of the "variance around the regression line" while in computing the ordinary correlation ratio one uses a numerator consiting of the actual variance around array means.[21] In part the distinction between the two is blurred when working with ordinal coefficients because they are not sensitive to linearities of regression. Nevertheless, the manner in which the ordinary correlation ratio is defined seems

[21] If one instead defines the unexplained variance as the weighted average array "variance-analogue," one obtains Goodman and Kruskal's tau. Thus their tau appears to be the appropriate analogue of the correlation ratio when working, as here, with pair-scores of ± 1 or 0. See Somers, *op. cit.*

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to lend justification to our earlier suggestion that in certain situations one may want, in effect, to run the percentages in the direction of the dependent variable by computing d_{yx} when X is taken as dependent.

A NOTE ON TESTS OF SIGNIFICANCE FOR d_{xy} AND d_{yx}

As was noted at the outset, this paper is primarily concerned with descriptive rather than inferential statistics. However, a remark may be made on the testing of significance of these new coefficients.

It was pointed out above that d_{xy} and d_{yx}, like tau-b and gamma, employ Kendall's S in the numerator. Clearly, if the hypothesis that S=0 in the population is accepted, then the hypotheses that $d_{yx}=d_{xy}$ =tau-b=gamma=0 are also accepted. Consequently, a test for the significance of Kendall's S is equivalent to a test for the significance of any one of these coefficients, so long as the null hypothesis is that the population value of the coefficient is zero. The best discussion to date of this test, where many ties are present as in the usual contingency table situation, is Burr.[22] He concludes that in a wide class of situations, where the population is statistically independent, the sampling distribution of Kendall's S has mean zero and variance as presented by Kendall.

Such a test is like the usual χ^2 test in that it is testing the hypothesis of a statistically independent population distribution. But, as noted above, it is more powerful than χ^2 against alternative hypotheses of monotonic correlation.

It would of course be much more desirable to have a method of interval estimation, such as the method of confidence intervals, for the population value of d_{xy} and d_{yx}. Such a technique requires a knowledge of the sampling distribution of these statistics under "non-null hypotheses" about the population value, however, and this is as yet unknown. For tau, a limited theory is available that applies whether or not ties are present among the rankings,[23] and for gamma the theory is completely developed.[24]

SUMMARY

This paper has reviewed some current methods of statistical description in situations where ordinal measurement obtains. It has introduced a pair of coefficients which from a link between two main trends in earlier work in this field, that of Kendall, and that of Goodman and Kruskal. The new coefficients are shown to be equal to the percentage difference in fourfold tables, and a function of all possible percentage differences in at least special types of tables of larger size. In any case, the coefficients have an operational interpretation, are asymmetric, and may be applied to contingency tables of any dimension, so long as the categories have a natural ordering, and one wishes to measure monotonic correlation.

[22] Burr, *op. cit.*

[23] Kendall, *op. cit.*, p. 60.

[24] Leo A. Goodman and William G. Kruskal, "Measures of Association for Cross Classification, III," to be published.

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